

such that a small change in angle of attack produces a change of opposite sign in the pitching moment about the center of gravity. The mathematical criterion for pitch stability is then

$$\frac{\partial m}{\partial \alpha} = \frac{1}{2} \rho V_\infty^2 S_w \bar{c} \frac{\partial C_m}{\partial \alpha} < 0 \quad (4.2.7)$$

Since the dynamic pressure, the wing area, and the mean chord length are always positive, pitch stability requires that

$$\frac{\partial C_m}{\partial \alpha} \equiv C_{m,\alpha} < 0 \quad (4.2.8)$$

which is **the general pitch stability criterion for any aircraft**. For this reason, the derivative in Eq. (4.2.8) is usually called the *pitch stability derivative* or *pitch stiffness*. The word “stiffness” comes from an analogy with a torsional spring, which produces a restoring moment that is proportional to the angular displacement from equilibrium.

Using Eq. (4.2.5) in Eq. (4.2.8) gives

$$\frac{\partial C_{m_{ac}}}{\partial \alpha} - \frac{l_w}{\bar{c}} \frac{\partial C_L}{\partial \alpha} < 0 \quad (4.2.9)$$

By definition, the aerodynamic center is a point on the wing where the change in pitching moment coefficient with respect to angle of attack is zero,

$$\frac{\partial C_{m_{ac}}}{\partial \alpha} = 0 \quad (4.2.10)$$

and thus pitch stability requires

$$-\frac{l_w}{\bar{c}} \frac{\partial C_L}{\partial \alpha} < 0 \quad (4.2.11)$$

Since the change in lift coefficient with angle of attack is positive for angles of attack less than stall, we see that stability requires $l_w > 0$. This means that for static stability, the aerodynamic center must be aft of the center of gravity. This condition is opposite to that required for static equilibrium. Thus, **a simple cambered wing is not statically stable in free flight**.

Using the trim requirement in Eq. (4.2.6) to eliminate the distance l_w from the stability requirement in Eq. (4.2.11) gives

$$-\frac{C_{m_{ac}}}{C_L} \frac{\partial C_L}{\partial \alpha} < 0 \quad (4.2.12)$$

This is the requirement for stable trim in this tailless flying wing. The lift coefficient must be positive to support the weight of the aircraft and the lift slope is positive for any